Total Printed Pages: 03

Roll No. :.....

MAB-101
Examination – June - 2023
B.Tech. I Sem : Common for all Branches
Linear Algebra and Calculus

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & Part b – Objective type/Fill in the blanks/ One sentence answer,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) (I) Maclaurin's Series of e^x is 01

(i) $1 + x + {}^{x^2}/_{L^2} + {}^{x^3}/_{L^3} + \ldots\ldots\ldots$ (ii) $1 - x + {}^{x^2}/_{L^2} - {}^{x^3}/_{L^3} + \ldots\ldots\ldots$

(iii) $1 - x - {}^{x^2}/_{L^2} - {}^{x^3}/_{L^3} - \ldots\ldots\ldots$ (iv) $1 + x + {}^{x^2}/_{2} + {}^{x^3}/_{3} - \ldots\ldots\ldots$

(II) Taylor's Theorem of (fx) about $x = a$ is 01

(i) $f(a+h) = f(a) + hf'(a) + {}^{h^2}/_{L^2} f''(a) + \ldots\ldots$

(ii) $f(a+h) = f(a) + hf'(a) + {}^{h^2}/_{2} f''(a) + \ldots\ldots$

(iii) $f(a+h) = f(a) - hf'(a) + {}^{h^2}/_{L^2} f''(a) - \ldots\ldots$

(iv) $f(a+h) = f(0) - hf'(0) + {}^{h^2}/_{L^2} f''(0) - \ldots\ldots$

(b) Write the statement of Taylor's Theorem. 02

(c) Expand $Sinx$ in powers of x by Maclaurin's Theorem. 03

OR

Expand e^x in powers of $(x+3)$ by Taylor's Theorem. 03

(d) Prove that the radius of curvature at any point t of the curve $x = 3a\,Cost - a\,Cos3t$, $y = 3aSint - a\,Sint3t$ is $3a\,Sint$. 05

OR

Examine $f(x,y) = x^3 + y^3 - 3xy$ for maximum and minimum values. 05

Q.2 (a) If $u = e^{xyz}$ find the value of ${}^{\partial^3 u}/_{\partial x \partial y \partial z}$. 02

02

(b) What are the types of error?

03

(c) If $u = log_e \frac{x^4+y^4}{x+y}$ Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$.

OR

03

If $u = (1 - 2xy + y^2)^{-\frac{1}{2}}$ Prove that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = y^2u^3$

05

(d) If $u = u(y - z, z - x, x - y)$ Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

OR

05

If $u = xlogxy$ where $x^3+y^3 + 3xy = 1$, Find du/dx

Q.3 (a) Let $f(x) = \int_x^5 3t^2\, dt$, then $f(x)$ equals. 02

(i) $1/81$ (ii) $1/35$ (iii) $-1/81$ (iv) None of them

(b) The Limit when $n \to \infty$ of the series $\sum_{r=1}^{n-1} \frac{1}{\sqrt{n^2-r^2}}$ is. 02

(i) $2/\pi$ (ii) $1/2$ (iii) $\pi/2$ (iv) π

(c) Determine by integration the limit to which the sum of the following series tends as $n \to \infty$ 03

$$\frac{n}{n^2} + \frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \cdots + \frac{n}{n^2+(n-1)^2}$$

OR

Evaluate

03

$$\int_0^2 \frac{x^4}{\sqrt{4-x^2}}\, dx$$

(d) Evaluate

05

$$\iint_R xydxdy$$

Where R is the quadrant of the circle $x^2 + y^2 = a^2$ where $x \geq 0$ and $y \geq 0$.

OR

Evaluate

05

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dydx}{1+x^2+y^2}$$

Q.4 (a) Define the Rank of a Matrix. 02

(b) State Cayley-Hamilton Theorem for the matrix 02

(c) If 03

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$ Find A^{-1}

OR

Reduce the matrix A into Diagonal form 03

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

(d) Investigate for what value of λ and μ the system of equations 05

$$2x + 3y + 5x = 9$$
$$7x + 3y - 2z = 8$$
$$2x + 3y - \lambda z = \mu$$

Have (i) No solution (ii) A Unique solution (iii) An infinite number of solution

OR

Find the Eigen Values of the matrix 05

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q.5 (a) Define the following:

(i) Walk (ii) Path (iii) Circuit 02

(b) Show that the maximum number of edges in a simple graph with n vertices are $n(n-1)/2$. 02

(c) State and prove the De'Morgan Law for any statements p and q. 03

OR

Construct the truth table for the proposition $(p \wedge q) \Rightarrow (p \vee q)$. 03

(d) What are the Laws of the Algebra of propositions? 05

OR

Solve the following Boolean Expression: 05

$$x.y + \overline{x.z} + x.\bar{y}.z\,(x.y + z)$$

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